ROBOTAKSİ-BİNEK OTONOM ARAÇ YARIŞMASI

HAZIR ARAÇ MİMARİ TANIMLAMA DOKÜMANI

TEKNOFEST
HAVACILIK, UZAY VE TEKNOLOJİ FESTİVALİ

1 İçindekiler

2 Donanım Mimarisi

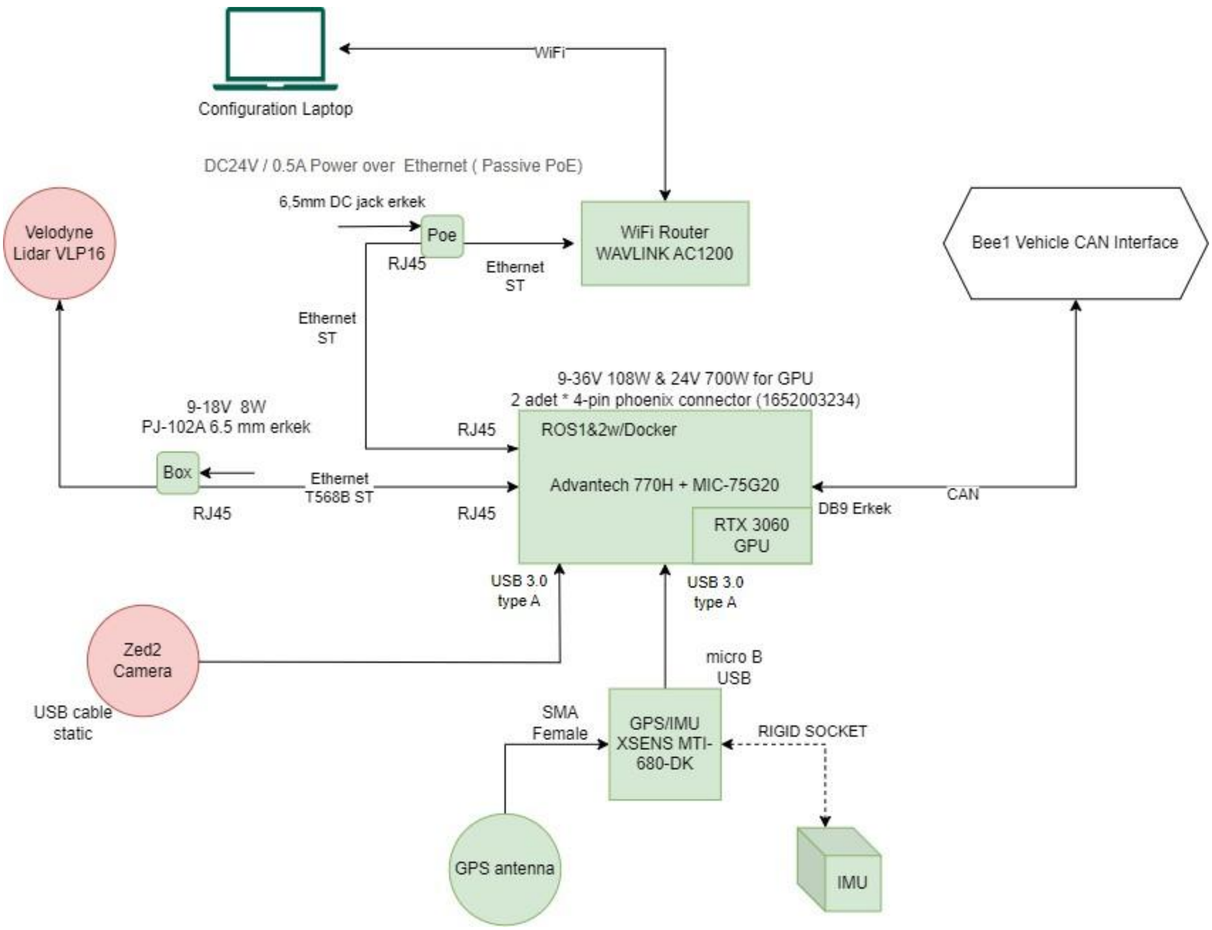


Figure 1 Donanım Mimarisi

Bileşenler

- İşlemci Birimi
- Lidar
- Stereo Kamera
- GPS/IMU Alıcı
- WiFi Router

3 Bileşenler

3.1 İşlemci Birimi

Figure 2 Advantech MIC-770V3H Endüstriyel Box

3.1.1 Advantech MIC-770V3H Endüstriyel Box PC

(w/ MIC-75G20 GPU Expansion Module for Video AI)

Intel Core i9-12900TE İşlemci, 32GB RAM, 1TB SSD, RTX-3060 GPU Kartı

Features

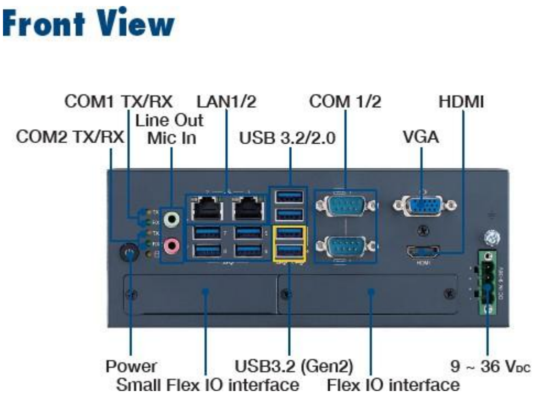
MIC-770 V3

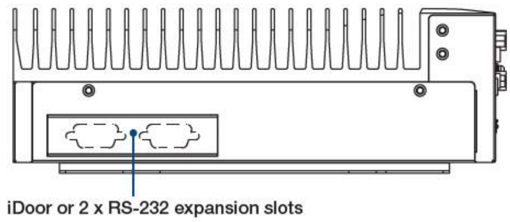
- Intel® 12th/13th Gen Core™ i CPU socket-type (LGA1700) with Intel® R680E/
- H610E chipset
- Wide operating temperature (-20 ~ 60 °C)
- VGA and HDMI output
- 2 x GigaLAN, 2 x USB 3.2 (Gen2) and 6 x USB 3.2 (Gen1)
- 2 x RS-232/422/485 and 4 x RS232 serial ports (Optional)
- 1 x 2.5" HDD/SSD, 1 x mSATA, and 1 x NVMe M.2
- 9 ~ 36 VDC input power range

- IP40 dust proof for deployment in harsh environment
- Supports FlexIO and iDoor technology, flexible configure additional HDMI,
- DP, DVI, COM port, DIO, Remote switch IO
- Supports Advantech i-Modules
- Supports Advantech SUSIAPI and embedded software APIs
- Supports Intel® vPro™/AMT and TPM technologies
- Supports Advantech iBMC 1.2 remote out-of-band power management
- solution on DeviceOn

with i9-12900TE processor

- Core/Thread number 16/24
- Base Frequency 1.10 GHz
- Max Turbo Frequency 4.80 GHz
- L3 Cache 30 MB
- Temperature -20 ~ 60 °C
- Chipset R680E/H610E
- BIOS AMI 256Mb/128Mb SPI Flash





3.1.2 MIC-75G20 GPU Expansion Module for Video AI

Edge Computing with MIC-7 Series

- Expansion slot Slot 1/2: Blank, Slot 3: PCIe x16, Slot 4: PCIe x4
- SATA Connector 1 x SATA Signal, 1 x SATA Power
- Storage 2 x 2.5" swappable HDD/SSD storage bay
- Power
- Input: Dual 24 VDC (one on MIC-7000 system, one on MIC-75G20)

- Power consumption: Max. Load: 448W (tested with 350W GPU, PoE card and MIC-770 system with 65W CPU)
- Power solution supports up to maximum 700W (Tested with 350W GPU card's peak power consumption)
- x 6-pin Conn. for GPU card (12VDC, 17A for each Conn.)
- 1 x 4-pin Conn. for add-on card (12VDC, 5A)
- GPU Card Dimension Thickness: 59.38 mm (2.75-slot width), Length: 334.6 mm, Height: 110.9 mm
- Support up to triple-fan fan GPU cards
- LED 1x indicating LED for power status
- Enviroment
- Operating Temp.: -10~60 °C (35W CPU w/ industrial wide-temp. RAM/SSD)
- Vibration: With SSD: 1 Grms @ 5~500 Hz, randon, 1 hr/axis
- Shock: With SSD: 10G, IEC-68-2-27, half-sine wave, 11 ms duration
- Mechanical
- MIC-75G20 N.W. 3.5 kg; G.W.: 5.5 kg
- Dimension (W x H x D): 130 x192 x 385 mm
- Fan 1x 12025 cooling fan embedded (2200 RPM, 82 CFM, Max. 36.5 dB)

Figure 3 MIC-75G20

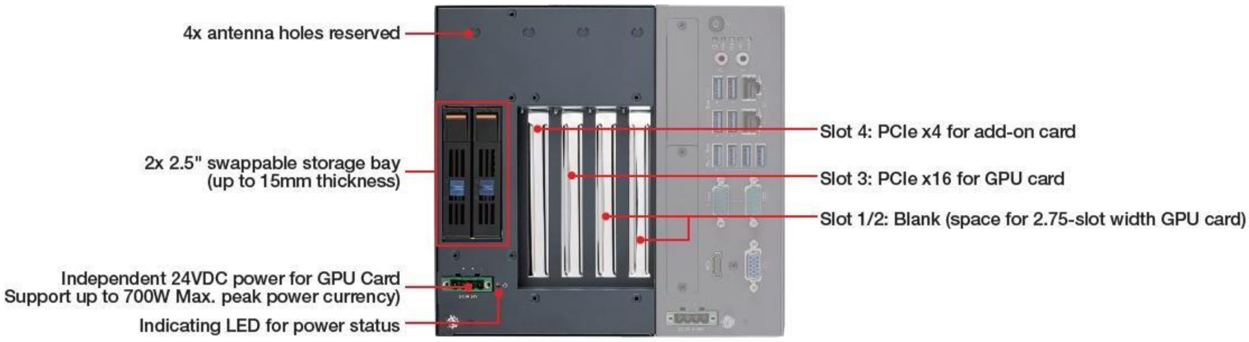


Figure 4 MIC-75G20 Slots

3.1.3 RTX-3060 GPU Kartı (12GB)

NVIDIA GeForce RTX 3060 GPU

• 3584 CUDA cores

• 12GB GDDR6 memory

• 192-bit memory bus • Engine boost clock: 1777 MHz

• Memory clock: 15.0 Gbps

• **PCI Express 4.0 16x**

600-watt power supply recommended (170-watt max power consumption • 8-pin power connector)

3.2 Lidar

Velodyne VLP-16 (take over from previous platform)

• Time of flight distance measurement with calibrated reflectivities

• 16 channels • Measurement range up to 100 meters

• Accuracy: +/- 3 cm (typical)

• Dual returns

• Field of view (vertical): 30° (+15° to -15°)

• Angular resolution (vertical): 2°

• Field of view (horizontal/azimuth): 360°

• Angular resolution (horizontal/azimuth): 0.1° - 0.4°

• Rotation rate: 5 - 20 Hz

• Integrated web server for easy monitoring and configuration

Figure 5 VLP-16

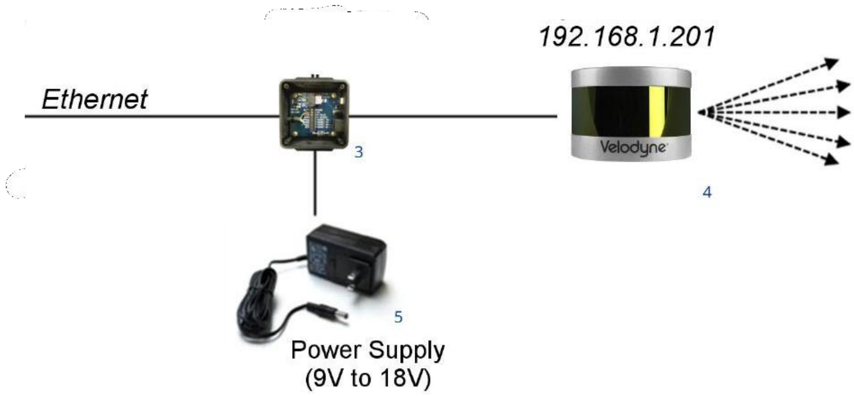


Figure 6 VLP-16 Connection

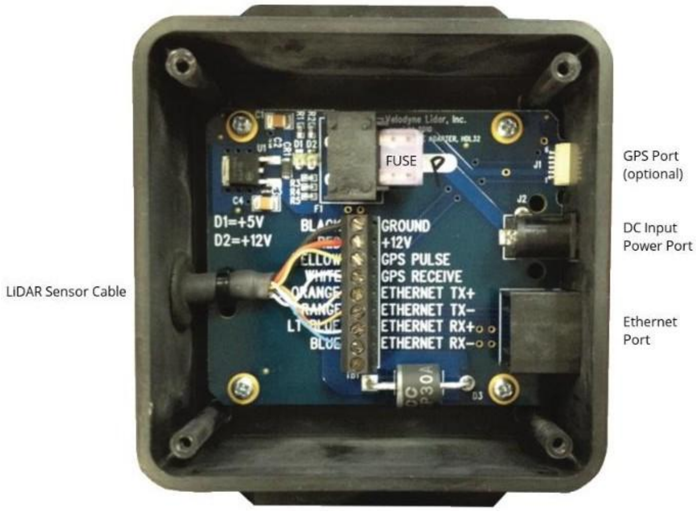


Figure 7 Distribution Box connectors

3.3 Stereo Kamera

((take over from previous platform)

Figure 8 ZED2 Camera

Camera	
Output Resolution	2x (2208x1242) @15fps 2x (1920x1080) @30fps 2x (1280x720) @60fps 2x (672x376) @100fps
Field of View	Max. 110°(H) x 70°(V) x 120°(D)
Interface	USB 3.0/2.0 Integrated 1.2m cable (3.97ft)
Depth Range	0.3 m to 20 m (0.98ft to 65.61ft)
Depth Accuracy	< 1% up to 3m (9.84ft) < 5% up to 15m (49.21ft)

Sensors	
Motion	Gyroscope, Accelerometer, Magnetometer
Environmental	Barometer, Temperature

Physical	
Dimensions	174.9 x 29.8 x 31.9mm (6.89 x 1.18 x 1.25")
Weight	164g (0.36 lb.)
Op. Temp.	-10 °C to +45°C (14°F to 113°F)
Power	380 mA / 5V USB Powered

Figure 9 ZED2 Specifications

3.4 GPS/IMU Alıcı

MTI-680-DK XSENS

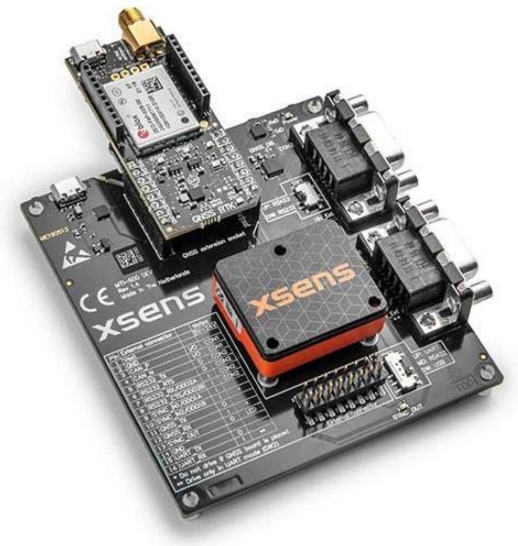


Figure 10 MTI-680-DK XSENS

Horizontal pos. accuracy PVT [m] 1.5 (Without RTK)

Supported GNSS signals: GPS L1C/A [MHz] 1575.42 L2C [MHz] 1227.60 GLONASS L1OF [MHz] 1598.0 - 1605.4 L2OF [MHz] 1242.9 – 1248.6 Galileo E1-B/C [MHz] 1575.42 E5b [MHz] 1207.14 BeiDou B1I [MHz] 1561.098 B2I [MHz] 1207.14

Table 6: MTi 600-series gyroscope specifications

Gyroscope specification[4]	Unit	Value
Standard full range	[°/s]	±2000
In-run bias stability	[°/h]	8
Bandwidth (-3dB)	[Hz]	520
Noise density	[°/s/√Hz]	0.007
g-sensitivity (calibrated)	[°/s/g]	0.001
Non-linearity	[%FS]	0.1
Scale Factor variation	[%]	0.5 (typical) 1.5 (over life)

Table 7: MTi 600-series accelerometer specifications

Accelerometer[4]	Unit	Value
Standard full range	[g]	±10
In-run bias stability (x, y)	[µg]	10
In-run bias stability (z)	[µg]	15
Bandwidth (-3dB)	[Hz]	500
Noise density	[µg/√Hz]	60
Non-linearity	[%FS]	0.1

Table 8: MTi 600-series magnetometer specifications

Magnetometer[4]	Unit	Value
Standard full range	[G]	±8
Non-linearity	[%]	0.2
Total RMS noise	[mG]	1
Resolution	[mG]	0.25

Table 9: MTi 600-series barometer specifications

Barometer[4]	Unit	Value
Full range	[hPa]	300-1250
Total RMS Noise	[Pa]	1.2
Relative accuracy	[Pa]	±8 [5]

Figure 11 Sensor Specifications

*The micro USB port on the main board can be used to connect the MTi-600 to a host though the included micro USB cable.

3.5 WiFi Router

AERIAL HD4 – AC1200 Dual-band High Power Outdoor Wireless AP/Range Extender/Router with PoE and High Gain Antennas

Figure 12 WAVLINK AC1200

Ports

1 x 10/100/1000Mbps WAN

1 x 10/100Mbps LAN

Power Supply

DC24V / 0.5A Power over Ethernet (Passive PoE)

WIRELESS

Standard

IEEE 802.11a/b/g/n/ac

Speed

IEEE 802.11ac: up to 867Mbps IEEE 802.11n: up to 300Mbps IEEE 802.11g: 54Mbps IEEE 802.11b: 11Mbps

IEEE 802.11a: up to 54Mbps

Frequency band

2.4GHz, 5GHz

Wireless Security

WPA-PSK/ WPA2-PSK encryption

Antennas

4 x 7dBi Detachable Omni Directional Antennas

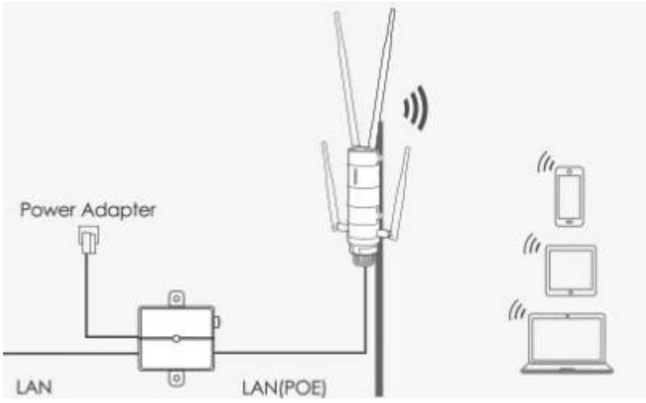


Figure 13 Network Setup

4 Yazılım Mimarisi

ROTAXI işlemci birimi (Advantech) üzerinde Linux Ubuntu 20.04 işletim sistemi çalışmaktadır. Linux işletim sistemin üzerinde ROS1 ve ROS2 ortamları kurulu olarak gelecektir. Kolay kurulum ve uyumluluk problemlerini aşmak amacıyla yazılımlar Docker ortamında çalıştırılacaktır. ROS1 veya ROS2 ortamları seçimli olarak başlatılabilmektedir.

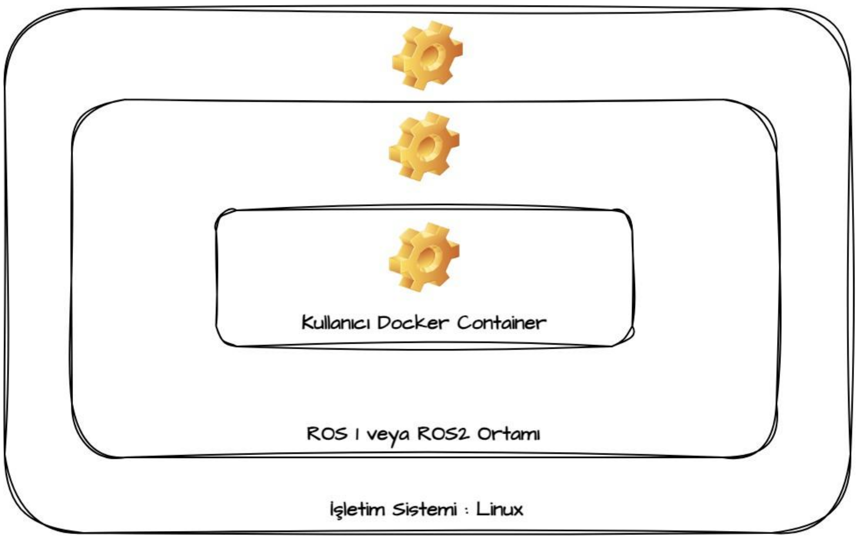


Figure 14 Yazılım Katmanları

Kullanılan ortamların versiyon bilgisi aşağıda verilmiştir.

- İşletim Sistemi: Linux Ubuntu 20.04 LTS (focal forsa)
- Kütüphane: ROS1 (noetic) & ROS2 (foxy)

- Sanal Ortam: Docker

Sistem ROS1 veya ROS2 olarak ayağa kaldırıldığında gerekli kontrol ve iletişim nodeları başlatılmaktadır. Bu ROS nodeları vasıtasıyla kullanıcı araca yüksek seviye ROS mesajları ile komut gönderebilmekte ve araç bilgilerini yine ROS mesajları üzerinden okuyabilmektedir. Genel mesajlaşma mimarisi aşağıda gösterilmiştir.

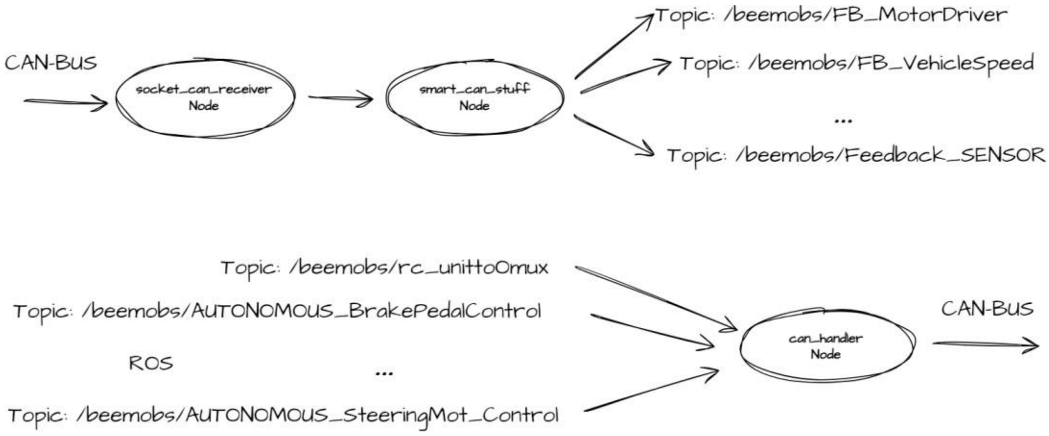


Figure 15 ROS node mimarisi

Ayrıca araç üstündeki sensor verilerine ROS ortamından ulaşmak mümkündür. Lidar point cloud bilgisi, stereo kamera RGB görüntü ve derinlik haritası ayrıca GPS/IMU sensörü üzerinden konum ve kinematic bilgilere ulaşılarak otonom sürüş için gerekli altyapı sunulmaktadır. Sensörler aşağıdaki diagramda gösterilmiştir.

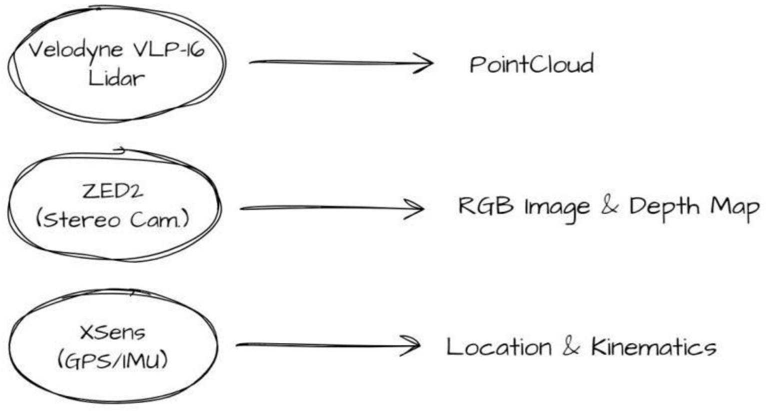


Figure 16 Sensörler